

Systematic search: “Spanish plume” in the meteorological literature

Scope: English-language, peer-reviewed meteorological journal articles published **on or before 13 August 2023** containing the exact phrase “Spanish plume.”
Databases specified: Google Scholar · Wiley / Royal Meteorological Society · AMS Journals Online · Web of Science
Search conducted: **2 August 2026** · Model: **Claude Opus 5 (Claude Code, live database access)**

1. Important limitation, up front

Two of the four required databases could not be searched from this session. This materially limits recall, so the yield below should be treated as a floor, not a complete census.

Database	Query	Status	Result count
Google Scholar	"Spanish plume" , custom range end-year 2023 (as_yhi=2023)	Partial. Retrieved 15 unique records across result pages 1–2, then Google served a bot-detection interstitial. Not bypassed.	Reported “About 43”; pagination later reported “18”. Scholar’s estimates are unstable and contradicted each other. 15 records actually retrieved.
Wiley Online Library / RMetS portal (rmets.onlinelibrary.wiley.com)	AllField="Spanish plume" , all years, Articles & Chapters	Complete. All results retrieved and enumerated.	76
AMS Journals Online (journals.ametsoc.org)	"Spanish plume" , all journals	Blocked. CloudFront WAF returned HTTP 403 to both direct fetch and the automated browser — for the search endpoint <i>and</i> article pages.	No count obtainable
Web of Science	Topic / full-text "Spanish plume"	Blocked. Navigation denied; also requires institutional authentication unavailable to this session.	No count obtainable

To partially compensate, two accessible channels were substituted, clearly marked as substitutes:

Substitute source	Query	Result count
OpenAlex full-text index (for Web of Science)	filter=fulltext.search:"spanish plume"	62
Site-restricted web search (for AMS)	"Spanish plume" restricted to journals.ametsoc.org , three query variants	6 unique AMS records

Independence from Schultz et al. Neither Schultz et al.’s review nor its dataset was consulted to identify any paper. Both Schultz et al. papers surfaced as *search hits* — unavoidable, as they are the top-ranked results for this phrase — and both postdate the cutoff, so they are excluded on date. Every paper listed below was found through independent database querying.

2. Totals

- **Records retrieved across all sources: 159** (15 Scholar + 76 Wiley + 62 OpenAlex + 6 AMS)
- **Unique records after deduplication: 130**
- **Eligible after screening: 15 confirmed**, plus ~35 probable-but-unverifiable

3. A second limitation: paywalls block reference-list screening

The exclusion criterion “papers where ‘Spanish plume’ appears only in the reference list” requires reading body text. Body text was verified by fetching each article’s full text and locating the phrase in context, *with the bibliography stripped out first*.

This worked only for open-access articles. Roughly **30 paywalled Wiley articles returned HTTP 403 on the full text while still exposing their reference lists** — and several of those reference lists do contain the phrase (they cite Morris 1986 or Webb & Pike 1998). Those are precisely the ambiguous cases the criterion targets, and they could not be resolved.

This is also *why* Wiley returned 76 hits: its search engine indexes bibliographies and does not enforce strict phrase matching.

4. Eligible papers — body-text mention confirmed

All 15 verified by locating the exact phrase in main text with the reference list excluded. Sorted by year.

#	Title	Authors	Year	Journal	DOI	Database(s)
1	Thunderstorms and hail on 7 June 1996: An early season ‘Spanish plume’ event	Webb, J.D.C.; Pike, W.S.	1998	Weather 53(8)	10.1002/j.1477-8696.1998.tb06391.x	Wiley; OpenAlex
2	Thunderstorms from a Spanish Plume event on 28 June 2011	Sibley, A.	2012	Weather 67(6)	10.1002/wea.1928	Wiley; OpenAlex
3	A climatology of convective available potential energy in Great Britain	Holley, D.M.; Dorling, S.R.; Steele, C.J.; Earl, N.	2014	Int. J. Climatology 34(14)	10.1002/joc.3976	Wiley; OpenAlex
4	Characterisation of convective regimes over the British Isles	Flack, D.L.A.; Plant, R.S.; Gray, S.L.; Lean, H.W.; Keil, C.; Craig, G.C.	2016	Q. J. R. Meteorol. Soc. 142(696)	10.1002/qj.2758	Wiley; Scholar; OpenAlex
5	An objective verification system for thunderstorm risk forecasts	Brown, K.; Buchanan, P.	2018	Meteorological Applications 26(1)	10.1002/met.1748	Wiley; OpenAlex
6	Investigation of the temporal variability of thunderstorms in central and western Europe and the relation to large-scale flow and teleconnection patterns	Piper, D.A.; Kunz, M.; Allen, J.T.; Mohr, S.	2019	Q. J. R. Meteorol. Soc. 145(725)	10.1002/qj.3647	Wiley; OpenAlex
7	Ambient conditions prevailing during hail events in central Europe	Kunz, M.; Wandel, J.; Fluck, E.; Baumstark, S.; Mohr, S.; Schemm, S.	2020	Nat. Hazards Earth Syst. Sci. 20	10.5194/nhess-20-1867-2020	OpenAlex
8	Comparison of short-period daytime convective rainfall accumulations with total column precipitable water	Young, M.; Lamb, D.; Lane, A.; Lattimore, C.	2020	Meteorological Applications 27(2)	10.1002/met.1903	Wiley; Scholar; OpenAlex
9	Severe thunderstorms with large hail across Germany in June 2019	Wilhelm, J.; Mohr, S.; Punge, H.J.; Mühr, B.; Schmidberger, M.; Daniell,	2021	Weather 76(7)	10.1002/wea.3886	Wiley

#	Title	Authors	Year	Journal	DOI	Database(s)
		J.E.; Bedka, K.M.; Kunz, M.				
10	Two hundred years of thunderstorms in Oxford	Burt, S.	2021	Weather 76(7)	10.1002/wea.3884	Wiley; Scholar; OpenAlex
11	North Atlantic air pressure and temperature conditions associated with heavy rainfall in Great Britain	Barnes, A.P.; Svensson, C.; Kjeldsen, T.R.	2021	Int. J. Climatology 42(5)	10.1002/joc.7414	Wiley
12	A characterisation of Alpine mesocyclone occurrence	Feldmann, M.; Germann, U.; Gabella, M.; Berne, A.	2021	Weather and Climate Dynamics 2	10.5194/wcd-2-1225-2021	OpenAlex
13	A regional lightning climatology of the UK and Ireland and sensitivity to alternative detection networks	Hayward, L.; Whitworth, M.; Pepin, N.; Dorling, S.	2022	Int. J. Climatology 42(15)	10.1002/joc.7680	Wiley; Scholar; OpenAlex
14	Thunderstorm tracking in Northwest Europe for enhanced hazard preparedness	Hayward, L.; Whitworth, M.; Pepin, N.; Dorling, S.	2023	Int. J. Climatology 43(11)	10.1002/joc.8123	Wiley; OpenAlex
15	Meteotsunami in the United Kingdom: the hidden hazard	Lewis, C.; Smyth, T.; Williams, D.; Neumann, J.; Cloke, H.	2023	Nat. Hazards Earth Syst. Sci. 23	10.5194/nhess-23-2531-2023	OpenAlex

Both 2023 entries clear the cutoff on online-publication date (30 May and 17 July 2023 respectively). DOIs resolve at <https://doi.org/<DOI>>.

5. Probable but unverifiable (~35 records)

These passed topic, language, region and date screening and matched in at least one database, but their body text was paywalled, so it could not be confirmed that the mention is not reference-list-only. Notable examples:

- Collier & Lilley 1994 (*Meteorol. Appl.*); Done et al. 2006 and Uebel & Bott 2015 (*QJRMS*)
- Owens 2016 and Grahame et al. 2015 (*Weather*); Young & Grahame 2022–2023 Parts 2/3/6 (*Weather*)
- Mohr et al. 2019 and Clarke et al. 2019 Parts 1–2 (*QJRMS*); Stone et al. 2022 and Wilkinson & Neal 2021 (*QJRMS*)
- Allan et al. 2019 (*JJC*); Lewis & Gray 2010 (*Atmos. Res.* — the canonical UK MCS citation); van Delden 1998 (*Meteorol. Atmos. Phys.*)
- The AMS set: Dahl & Fischer 2016 (*WAF*); Schultz 2005 (*MWR*); Groenemeijer et al. 2017 (*BAMS*); Púčik et al. 2017 and Taszarek et al. 2020 (*J. Climate*)

Eight of these were strict-phrase Google Scholar hits, which confirms the phrase is present in the indexed full text but **not** that it falls outside the bibliography — Scholar indexes reference lists too.

6. Exclusion reasons

Reason	Approx. n	Examples
Not a peer-reviewed journal article	~30	Complete-issue records (<i>Weather</i> 77(7), 78(1), 78(3), 78(11)); “Issue Information”; indexes; Society News; Readers’ Forum; Letters to the Editor; a book review; conference abstracts; preprints; dissertations; an NWO report; a <i>Conversation</i> explainer

Reason	Approx. n	Examples
Non-meteorological (loose-match noise)	~20	Palgrave chapter on Mediterranean migration poetry; horehound plume moth thesis; <i>Gertrude of Wyoming</i> ; “Tagging and User-Contributed Metadata”; Lake Huron tritium plume; Valladolid urban CO ₂ /CH ₄ plume; Jupiter equatorial plumes; solar polar plumes
Published after 13 August 2023	~15	Both Schultz et al. papers (2025); Heathrow hazardous-weather synoptic analysis (24 Sep 2023); <i>Weather</i> 79(1) (Jan 2024); France derecho climatology (2024); atmospheric-desert papers (2024–2026); Saharan dust / hail <i>ACP</i> (2025)
Weather outside Europe	~4	Cumulonimbus convection over India; elevated mixed layers in South and North America (<i>MWR</i> 2018)
Non-English	2	Piper’s KIT dissertation (German); León WRF / radiometer thesis (Spanish)
Phrase absent on inspection (database false positive)	≥1 confirmed	<i>NHESS</i> 17, 1319 (2017), lightning / NAO — full text fetched and the phrase does not occur. Given Wiley’s loose matching, more of the 76 are likely in this category
Reference-list-only	Unresolved	Several paywalled Wiley records show the phrase in their bibliography with body text inaccessible — could not be adjudicated

7. What would close the gaps

An authenticated AMS Journals and Web of Science session, plus institutional Wiley access, would resolve both the two missing databases and the ~35 unverifiable records. Without them, the 15 confirmed papers are solid, but recall is incomplete — most obviously across AMS titles (*MWR*, *WAF*, *BAMS*, *J. Climate*), where only six records were visible, and then only through a search engine rather than the database itself.